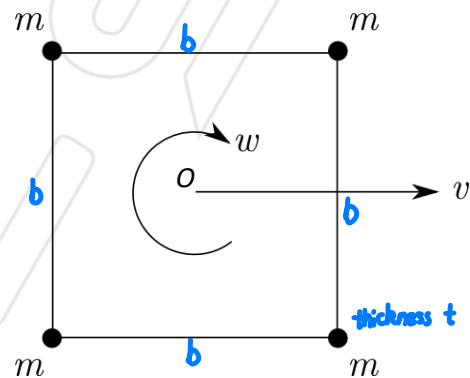


AERO40007 Progress Test 1

NAME	CID	GRADE (/100)

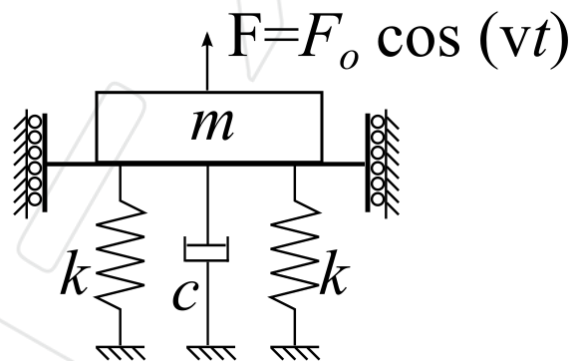
You can use the Datasheet available in Blackboard.

- The figure shows a square body with side length b , thickness t and constant density ρ . Four point masses m are attached at the corners. The body has a constant horizontal velocity v and a constant angular velocity about the centre of mass ω .



What is the total kinetic energy of the body?

- Write the equation of motion. Calculate the natural frequency and damping ratio of the system below:



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60/100

1. $KE = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$$\begin{aligned} I &= \sum mr^2 = \frac{1}{3}m[(\frac{1}{2}b)^2 + (\frac{1}{2}b)^2] + 4[2m(\frac{1}{2}b)^2 + (\frac{1}{2}b)^2 2m] \\ &= \frac{1}{3}\rho b^2 t (\frac{1}{2}b^2) + 4(mb^2) \\ &= \frac{1}{6}\rho b^4 t + 4mb^2 \end{aligned}$$

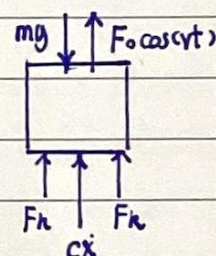
$$I = \sum mr^2 = \frac{1}{12}b \cdot b^3 \cdot \rho \cdot t + \frac{1}{12}b \cdot b^3 \cdot \rho \cdot t + 4(m \cdot (\frac{b}{2})^2) = \frac{1}{6}\rho b^4 t + 2mb^2$$

$$\begin{aligned} I &= \sum mr^2 = \frac{1}{12}b \cdot b^3 \cdot \rho \cdot t + 4[2m(\frac{1}{2}b)^2 + (\frac{1}{2}b)^2 2m] \\ &= \frac{1}{12}\rho b^4 t + 4mb^2 \end{aligned}$$

$$\begin{aligned} \therefore KE &= \frac{1}{2}(4m + b^2 t \rho) v^2 + \frac{1}{2}(\frac{1}{12}\rho b^4 t + 4mb^2) \omega^2 \\ &= 2mv^2 + \frac{1}{2}\rho b^2 t v^2 + \frac{1}{12}\rho b^4 t \omega^2 + 2mb^2 \omega^2 \end{aligned}$$

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2. $m\ddot{x} + c\dot{x} + kx = -F_x = -F_0 \cos(\omega t)$



$$\therefore x(t) = A \cos(\omega t - \phi)$$

$$A = \frac{F_0}{\sqrt{(2k - m\omega^2)^2 + (c\omega)^2}}$$

$$\therefore x(t) = \frac{F_0}{\sqrt{(2k - m\omega^2)^2 + (c\omega)^2}} \cos(\omega t - \phi)$$

$$\omega_n = \sqrt{\frac{2k}{m}}$$

$$\zeta = \frac{c}{2m\omega_n} = \frac{c}{2m} \sqrt{\frac{m}{2k}} = \frac{\sqrt{2}c}{4\sqrt{km}}$$

$$\therefore \text{ratio} = \frac{T_{\text{damped}}}{T_{\text{undamped}}} = \frac{1}{\sqrt{1 - \zeta^2}}$$

$$\therefore \sqrt{1 - \zeta^2} = \sqrt{1 - \frac{2kc^2}{16km}} = \sqrt{1 - \frac{kc^2}{8km}}$$

$$\therefore \text{ratio} = (1 - \frac{kc^2}{8km})^{-1/2}$$

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